Rigorous Execution of Semantic Layer Formalization and Falsifiable Hypotheses for Structural Aliasing in Time-Series Foundation Models

Coursework Project Deliverable — Sections 3 & 4
Joint Computer Security / Digital Transformation Graduate Program
Co-supervised with Prof. Benini

May 22, 2026

Abstract

This report delivers a full, production-grade formal execution of **Point 3 (Semantic Layer and Knowledge Representation)** and **Point 4 (Formulation of Falsifiable Hypotheses)** as specified in the coursework guidelines. Centered on the energy management domain and grounded in the 1-minute telemetry of the `SolarTechLab` dataset, we investigate the structural aliasing phenomenon native to patch-based time-series foundation models, with specific execution calibrated to Amazon's Chronos architecture. We construct a formal description-logic ontology mapping continuous multi-variate solar signals to high-level operational concepts, mathematically model the "between-patch" information leak, and establish three rigorous, falsifiable hypotheses with explicit testing criteria to isolate architectural token-space distortions from classical sampling limits.

1 Introduction and Theoretical Foundations

The deployment of large language models (LLMs) and transformer backbones for time-series forecasting has introduced a critical paradigm shift: continuous physical signals are mapped into discrete representation spaces. In frameworks such as Amazon's *Chronos*, this mapping is governed by two sequential operators: value-space quantization and temporal-space patching.

Let a continuous-time physical process be sampled uniformly at a frequency $f_s = \frac{1}{\Delta t}$. For the empirical baseline provided by the `SolarTechLab` dataset, the sampling interval is precisely $\Delta t = 1$ minute, yielding $f_s = 1\ \text{min}^{-1} \approx 0.0167$ Hz. The resulting discrete-time multi-variate signal is denoted as a sequence of vectors $\mathbf{x}_t \in \mathbb{R}^d$:

$$\mathbf{x}_t = \begin{bmatrix} P_{\text{PV}}(t) & T_{\text{air}}(t) & G_{\text{h}}(t) & G_{\text{tilt}}(t) & W_{\text{s}}(t) & W_{\text{d}}(t) \end{bmatrix}^T \tag{1}$$

where $P_{\text{PV}}(t)$ is the photovoltaic power production (kW), $T_{\text{air}}(t)$ is the ambient air temperature (°C), $G_{\text{h}}(t)$ and $G_{\text{tilt}}(t)$ represent the global horizontal and global tilted solar irradiances (W/m^2), and $W_{\text{s}}(t), W_{\text{d}}(t)$ capture wind speed and direction respectively.

The Chronos framework ingests a history window of these variables, applies a local scaling transformation (typically mean scaling), and quantizes the normalized continuous values into a fixed vocabulary $\mathcal{V} = \{v_1, v_2, \ldots, v_C\}$, where $C = 4096$ represents the vocabulary size. Patch-based tokenization operators $\mathcal{P}_{P,S}$ compress this high-resolution sequence by grouping P consecutive time steps into a unified token vector or a localized spatial patch, advancing across the time axis with a stride of S steps.

While computationally efficient, this architectural design creates *structural aliasing* [cite: 253]. This phenomenon is not induced by undersampling relative to the Nyquist-Shannon limit, but by the mathematical projection of localized high-frequency or boundary-locked signal components into invariant token trajectories, rendering them structurally unobservable to the model's internal attention mechanisms [cite: 257-261].

2 Point 3: Semantic Layer and Knowledge Representation

To formalize the link between raw mathematical signal streams and domain meaning, we define a semantic layer that captures the operational states of the solar array [cite: 262-264]. In agentic AI environments, this layer serves as the formal cognitive framework through which an automated agent monitors its surroundings, explains observed variances, and executes control loops [cite: 265].

2.1 Ontological Specification of the Solar Energy Domain

We formalize the semantic layer using an ontology represented by the tuple:

$$\mathcal{O} = \langle \mathcal{C}, \mathcal{R}, \mathcal{A} \rangle \tag{2}$$

where $\mathcal{C}$ is the set of concepts (classes), $\mathcal{R}$ is the set of binary relations (properties), and $\mathcal{A}$ represents the set of terminological axioms (TBox) defining domain logic rules.

2.1.1 Concept Hierarchy ($\mathcal{C}$)

We specify four primary operational concepts under the global top concept $\top$:

$$\begin{aligned} \text{OperationalState} &\sqsubseteq \top \\ \text{NormalGeneration} &\sqsubseteq \text{OperationalState} \\ \text{TransientPerturbation} &\sqsubseteq \text{OperationalState} \\ \text{SystemAnomaly} &\sqsubseteq \text{OperationalState} \end{aligned}$$

These macro-concepts are populated by highly localized domain states extracted from the physical realities of the `SolarTechLab` dataset:

1. **ClearSkyPeak** (ClearSkyPeak $\sqsubseteq$ NormalGeneration): Represents maximum operational generation under unobstructed solar exposure. Formally governed by high global tilted irradiance and steady-state derivative bounds:

$$\mathcal{A}_1 : \text{ClearSkyPeak} \equiv \text{NormalGeneration} \sqcap \exists G_{\text{tilt}}.(>700\,\text{W/m}^2) \sqcap \forall \left| \frac{\partial G_{\text{tilt}}}{\partial t} \right| .(\leq \epsilon_1) \tag{3}$$

2. **InverterClipping** (InverterClipping $\sqsubseteq$ NormalGeneration): A critical system-level saturation regime where the physical power output reaches its hardware limit $P_{\max}$ despite expanding irradiance:

$$\mathcal{A}_2 : \text{InverterClipping} \equiv \text{NormalGeneration} \sqcap \exists P_{\text{PV}}.(=P_{\max}) \sqcap \exists G_{\text{tilt}}.(>\gamma_{\text{threshold}}) \tag{4}$$

3. **MicroCloudShading** (MicroCloudShading $\sqsubseteq$ TransientPerturbation): High-frequency cloud transients that induce swift, high-amplitude, localized drops in solar irradiance and power output, while meteorological parameters such as ambient temperature remain stationary:

$$\mathcal{A}_3 : \text{MicroCloudShading} \equiv \text{TransientPerturbation} \sqcap \exists \left| \frac{\partial G_{\text{tilt}}}{\partial t} \right| .(>\alpha) \sqcap \exists \left| \frac{\partial P_{\text{PV}}}{\partial t} \right| .(>\beta) \sqcap \forall \left| \frac{\partial T_{\text{air}}}{\partial t} \right| .(\leq \epsilon_2) \tag{5}$$

4. **SensorMalfunction** (SensorMalfunction $\sqsubseteq$ SystemAnomaly): Characterized by severe out-of-bounds telemetry corruptions embedded within the reference dataset, specifically the placeholder flag value -999999.0 present in the T_{air} column:

$$\mathcal{A}_4 : \text{SensorMalfunction} \equiv \text{SystemAnomaly} \sqcap \exists T_{\text{air}}.(= -999999.0) \tag{6}$$

2.1.2 System Relations and Mapping Axioms ($\mathcal{R}$)

The operational dynamics between the telemetry streams and semantic states are constrained by object properties:

$$\text{drives} \subseteq \text{GlobalIrradiance} \times \text{PVPowerOutput} \tag{7}$$
$$\text{attenuates} \subseteq \text{TransientPerturbation} \times \text{PVPowerOutput} \tag{8}$$
$$\text{interprets} \subseteq \text{AgenticAI} \times \text{TokenTrajectory} \tag{9}$$
$$\text{mapsTo} \subseteq \text{TokenTrajectory} \times \text{OperationalState} \tag{10}$$

2.2 Signal-to-Token-to-Concept Semantic Mapping

Let $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_L\}$ be an extended sequence of observations. The architectural patch operator groups this sequence into localized patches $\mathbf{P}_i \in \mathbb{R}^{P \times d}$, which Chronos projects into a continuous hidden representation sequence $\mathbf{T} = (\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_N)$, where $\mathbf{e}_i \in \mathbb{R}^m$. The semantic layer implements an interpretation operator $\mathcal{M} : \mathbb{R}^m \to \mathcal{C}$ that assigns a distinct domain concept to each token $\mathbf{e}_i$. The semantic state of the physical plant is preserved if and only if:

$$\mathcal{M}(\mathbf{e}_i) \equiv \text{True Operational State over } [(i-1)S+1, (i-1)S+P] \tag{11}$$

2.3 The Mechanism of the "Between-Patch" Blind Spot

A profound structural failure in this architecture occurs when high-frequency domain concepts manifest directly on the temporal interface separating two consecutive token patches, creating a severe semantic information leak.

Consider a discrete *MicroCloudShading* transient event spanning a localized window of $\delta =$ 10 minutes, characterized by a deep, local parabolic drop in $G_{\text{tilt}}(t)$ and a matching drop in $P_{\text{PV}}(t)$. We define two competing configurations of this signal relative to the model's patch grid:

2.3.1 Case A: Interior Alignment (Observable State)

The cloud transient is entirely contained within the support of a single patch $\mathbf{P}_i$. The local mean-scaling operator inside Chronos isolates the high local variance:

$$\sigma^2_{\mathbf{P}_i} = \frac{1}{P} \sum_{t \in \mathbf{P}_i} (x_t - \mu_{\mathbf{P}_i})^2 \gg 0 \tag{12}$$

The quantization step maps the continuous values into highly polarized, lower-bound vocabulary bins $\{v_{\text{min}}, \ldots, v_{\text{mid}}\}$. The resulting token embedding $\mathbf{e}_i$ exhibits a high-magnitude structural signature, mapping successfully under the interpretation framework to the correct concept:

$$\mathcal{M}(\mathbf{e}_i) = \text{MicroCloudShading} \tag{13}$$

2.3.2 Case B: Boundary Alignment (The Blind Spot Collision)

The identical 10-minute micro-shading event is translated across the timeline such that it is centered exactly on the boundary grid interface $t = i \cdot S$. The event is split symmetrically: $\delta/2 = 5$ minutes occur at the terminal edge of patch $\mathbf{P}_i$, and $\delta/2 = 5$ minutes occur at the initial edge of patch $\mathbf{P}_{i+1}$.

Because the patch operator compresses the temporal distribution of each window independently, the localized signal variance is heavily diluted across both tokens. For patch $\mathbf{P}_i$, the final 5 minutes of rapid power drop are averaged against $P - 5$ minutes of highly stable, peak unperturbed generation. Reciprocally, for patch $\mathbf{P}_{i+1}$, the first 5 minutes of the cloud transient are averaged against $P - 5$ minutes of subsequent stable recovery.

Consequently, the local mean-scaling factors for both windows remain dominated by the ambient peak generation, causing the 5-minute deviations to be absorbed as minor, low-amplitude noise. The quantization step maps the entire sequence of both patches into steady, mid-to-high vocabulary bins. The extracted token embeddings satisfy:

$$\mathbf{e}_i \approx \mathbf{e}_{i+1} \approx \mathbf{e}_{\text{stable}} \tag{14}$$

The ontology interpreter evaluates these flattened tokens and yields a highly distorted classification:

$$\mathcal{M}(\mathbf{e}_i) = \mathcal{M}(\mathbf{e}_{i+1}) = \text{ClearSkyPeak} \sqcap \exists \text{Attenuation}_{\text{negligible}} \neq \text{MicroCloudShading} \tag{15}$$

The sharp, high-frequency physical event has completely leaked between the patches, leaving zero architectural trace in the token space.

2.4 Downstream Agentic AI Risks and Vulnerabilities

When Chronos operates within an autonomous Agentic AI loop managing cyber-physical grid infrastructure, this structural semantic distortion introduces severe operational liabilities [cite: 265, 285]:

1. **Explanation Breakdown**: The agent monitors the invariant sequence $(\mathbf{e}_i, \mathbf{e}_{i+1})$ and generates a highly confident, yet completely false, natural language explanation for human operators: *"The system is operating at optimal nominal capacity under clear skies with negligible thermal degradation."* It masks an active meteorological perturbation that is actively draining power capacity [cite: 286].

2. **Grid Decision Hazards**: Because the agent fails to identify the discrete *MicroCloudShading* states due to boundary-layer dilution, its internal state machine estimates the spinning reserves incorrectly. It fails to trigger fast-acting battery discharge cycles or dispatchable generator throttling. If multiple micro-clouds pass over the array at intervals matching the patch boundaries, the unobserved power drops accumulate, leading to local grid frequency instability and potential cyber-physical equipment trips.

3 Point 4: Formulation of Falsifiable Hypotheses

To transition from a conceptual critique to an empirical, falsifiable validation framework, we translate the structural aliasing phenomenon into three testable hypotheses [cite: 266-267, 276]. These hypotheses isolate architectural tokenization distortions from classical sampling limits.

Let the formal patch tokenization operator acting on a discrete single-variable signal $X(t)$ be denoted by $\Phi(X; P, S) = \{\mathbf{P}_i\}_{i=1}^{N}$, where each patch vector $\mathbf{P}_i = [X((i-1)S+1), \ldots, X((i-1)S+P)]^T \in \mathbb{R}^P$. The Chronos encoder transforms each patch vector into a low-dimensional hidden representation vector $\mathbf{E}_i \in \mathbb{R}^m$.

3.1 Hypothesis I: Frequency-Dependent Blind Spots and Token Degeneracy

Hypothesis 3.1. Let the baseline environmental sampling interval be fixed at $\Delta t = 1$ minute. For a periodic physical signal $X(t) = A\sin(2\pi f t + \phi)$, if the underlying frequency f matches an integer harmonic of the patch-induced sampling frequency $f_p = \frac{1}{S \cdot \Delta t}$, such that:

$$f = k \cdot f_p = \frac{k}{S} \quad \text{for } k \in \mathbb{Z}^+ \tag{16}$$

then successive patch vectors become structurally invariant or exhibit complete redundancy up to a constant scalar, inducing a total representation collapse in the model's token space:

$$\mathbf{P}_i = \mathbf{P}_{i+1} \implies \mathbf{E}_i = \mathbf{E}_{i+1}, \quad \forall i \in \{1, \ldots, N-1\} \tag{17}$$

3.1.1 Falsifiable Condition and Test Criterion

Under the condition $f = k/S$, the sequence of hidden states generated by Chronos will satisfy $\Delta\mathbf{E} = \|\mathbf{E}_i - \mathbf{E}_{i+1}\|_2 \leq \xi$, where $\xi \to 0$ represents numerical precision noise. The model's internal self-attention matrix will collapse to a uniform distribution across the temporal tokens, rendering the underlying frequency completely unobservable, even though the signal exists well within the classical Nyquist bound ($f < 0.5$ min^{-1}).

Falsification Rule: If controlled empirical probing of Chronos's internal encoder states reveals that the distance metric $\Delta\mathbf{E}$ varies dynamically and maps linearly to the input signal's amplitude when $f = k/S$ under a non-overlapping grid ($S = P$), this hypothesis is definitively falsified.

3.2 Hypothesis II: Sensitivity to Temporal Alignment and Phase-Locking Shifts

Hypothesis 3.2. The visibility and structural representation of compact, localized transient solar phenomena within the Chronos token space is non-monotonic and highly sensitive to the continuous temporal phase shift τ of the event relative to the fixed architectural patch boundaries.

3.2.1 Mathematical Formulation

Let a localized micro-shading solar drop be modeled as a compact boxcar perturbation:

$$h(t; \tau) = -B \cdot \mathbb{I}_{[t_0+\tau,\, t_0+\tau+\delta]}(t) \tag{18}$$

where $\delta < S \leq P$. Let $\mathbf{E}_i(\tau)$ define the sequence of internal representation vectors extracted from Chronos under phase offset τ, and let $\mathbf{E}_0$ represent the unperturbed steady-state nominal token vector. We define the token-space information visibility metric as an integrated Euclidean distance:

$$\mathcal{D}(\tau) = \sum_{i=1}^{N} \|\mathbf{E}_i(\tau) - \mathbf{E}_0\|_2^2 \tag{19}$$

3.2.2 Falsifiable Condition and Test Criterion

The architectural information visibility function $\mathcal{D}(\tau)$ is minimized when the continuous phase offset aligns the midpoint of the compact transient exactly with the patch interface boundary:

$$\tau^* \implies t_0 + \tau^* + \frac{\delta}{2} = i \cdot S, \quad \text{for } i \in \mathbb{Z}^+ \tag{20}$$

At τ^*, the distance metric will exhibit a statistically significant collapse:

$$\frac{\mathcal{D}(\tau^*)}{\mathcal{D}(\tau_{\text{center}})} < \alpha_{\text{threshold}} \tag{21}$$

where τ_{center} represents an interior-aligned event centered at $(i-1)S + P/2$.

Falsification Rule: If empirical measurement shows that the token distance $\mathcal{D}(\tau)$ remains stationary, or increases, as the transient event is shifted across the patch boundary interface, or if the model's internal representations are invariant to temporal phase translations, this hypothesis is rejected.

3.3 Hypothesis III: Parametric Scaling Laws of Patch Geometry

Hypothesis 3.3. Structural aliasing and token-space degeneracies are a direct function of tokenization hyperparameters (P, S) and are governed by the patch overlap ratio:

$$\rho = 1 - \frac{S}{P} \tag{22}$$

For a fixed signal frequency $f = k/P$ (which induces complete token degeneracy under standard non-overlapping regimes where $S = P, \rho = 0$), systematically scaling the stride S to satisfy an architectural anti-aliasing criterion:

$$S < \frac{1}{2f} = \frac{P}{2k} \implies \rho > 1 - \frac{1}{2k} \tag{23}$$

breaks the phase-locking degeneracy, creating a time-varying trajectory in the token space and restoring frequency observability.

3.3.1 Falsifiable Condition and Test Criterion

We formulate an information-theoretic probing metric using the Minimum Description Length (MDL) or representation entropy $H(\mathbf{E})$ of the hidden state sequence. For a stationary input frequency f, a continuous sweep of the stride parameter S will reveal a sharp, non-linear phase transition in representation entropy. The mutual information $I(\mathbf{E}; X)$ between Chronos's internal embedding vectors and the continuous raw signal will undergo a step-function increase as S drops below the critical threshold $S^* = \frac{1}{2f}$.

Falsification Rule: If representation entropy $H(\mathbf{E})$ and mutual information $I(\mathbf{E}; X)$ scale linearly across all values of S, or if structural aliasing persists uniformly regardless of the overlap ratio ρ, this geometric scaling law hypothesis is falsified.

4 Calibration and Synthetic Signal Blueprint

To execute the controlled experiments in Point 5, we design a synthetic parameter generator strictly calibrated to the operational statistics of the `SolarTechLab` database. The synthetic baseline mirrors the real-world diurnal curve and introduces controlled perturbations to isolate the architectural effects:

$$X_{\text{synth}}(t) = X_{\text{diurnal}}(t) + X_{\text{transient}}(t; f, \tau) + \eta(t) \tag{24}$$

where $X_{\text{diurnal}}(t) = A_0 \max\left(0, \sin\left(\frac{2\pi t}{1440}\right)\right)$ models the 24-hour diurnal solar cycle (1440 minutes), $X_{\text{transient}}$ implements the controlled test signals for testing Hypotheses I, II, and III, and $\eta(t) \sim \mathcal{N}(0, \sigma^2)$ injects baseline sensor noise modeled directly on the standard deviation of the clean clear-sky periods in the dataset. This mathematical framework provides the explicit foundational baseline required to execute the empirical testing and subsequent Bayesian inference.